EPROM - Erasable programmable Read only memory.

I2C - smallest pin.

Lab Experiment: 1 **Date: 19/01/2024**

STUDY OF ARM EVALUATION SYSTEM

Objective:

To study the ARM processor system and to describe the features of its architecture. Software tools Requirement: Keil software

Introduction to ARM7

The LPC2141/42/44/46/48 microcontrollers are based on a 16-bit/32-bit ARM7TDMI-S CPU with real-time emulation and embedded trace support, that combine microcontroller with embedded high-speed flash memory ranging from 32 kB to 512 kB. A 128-bit wide memory interface and unique accelerator architecture enable 32-bit code execution at the maximum clock rate. For critical code size applications, the alternative 16-bit Thumb mode reduces code by more than 30 % with a minimal performance penalty. Due to their tiny size and low power consumption, LPC2141/42/44/46/48 are ideal for applications where miniaturization is a key requirement, such as access control and point-of-sale. Serial communications interfaces ranging from a USB 2.0 Full-speed device, multiple UARTs, SPI, SSP to I2C-bus and on-chip SRAM of 8 kB up to 40 kB, make these devices very well suited for communication gateways and protocol converters, soft modems, voice recognition, and low end imaging, providing both large buffer size and high processing power. Various 32-bit timers, single or dual 10-bit ADC(s), 10-bit DAC, PWM channels and 45 fast GPIO lines with up to nine edge or level sensitive external interrupt pins make these microcontrollers suitable for industrial control and medical systems.

Features

- 16-bit/32-bit ARM7TDMI-S microcontroller in a tiny LQFP64 package.
- 8 kB to 40 kB of on-chip static RAM and 32 kB to 512 kB of on-chip flash memory.
- 128-bit wide interface/accelerator enables high-speed 60 MHz operation.
- In-System Programming/In-Application Programming (ISP/IAP) via on-chip boot loader software. Single flash sector or full chip erase in 400 ms and programming of 256 bytes in 1 ms.
- EmbeddedICE RT and Embedded Trace interfaces offer real-time debugging with the on-chip RealMonitor software and high-speed tracing of instruction execution.
- USB 2.0 Full-speed compliant device controller with 2 kB of endpoint RAM. In addition, the LPC2146/48 provides 8 kB of on-chip RAM accessible to USB by DMA.
- One or two (LPC2141/42 vs. LPC2144/46/48) 10-bit ADCs provide a total of 6/14 analog inputs, with conversion times as low as 2.44 μs per channel.
- Single 10-bit DAC provides variable analog output (LPC2142/44/46/48 only).
- Two 32-bit timers/external event counters (with four capture and four compare
- channels each), PWM unit (six outputs) and watchdog.
- Low power Real-Time Clock (RTC) with independent power and 32 kHz clock input.
- Multiple serial interfaces including two UARTs (16C550), two Fast I2C-bus (400 kbit/s),
- SPI and SSP with buffering and variable data length capabilities.
- Vectored Interrupt Controller (VIC) with configurable priorities and vector addresses.
- Up to 45 of 5 V tolerant fast general purpose I/O pins in a tiny LQFP64 package.
- Up to 21 external interrupt pins available.
- 60 MHz maximum CPU clock available from programmable on-chip PLL with settling time of 100 μs
- On-chip integrated oscillator operates with an external crystal from 1 MHz to 25 MHz.
- Power saving modes include Idle and Power-down.
- Individual enable/disable of peripheral functions as well as peripheral clock scaling for additional power optimization.
- Processor wake-up from Power-down mode via external interrupt or BOD.

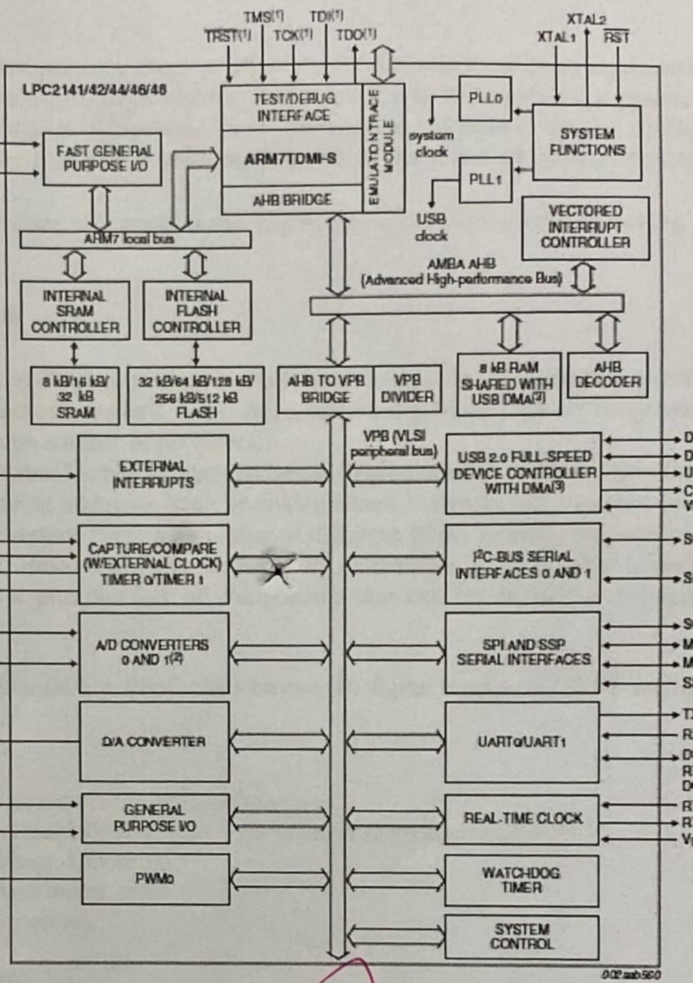
JTAG - Joint Test Action Group.

FRC - Flat Ribbon cable.

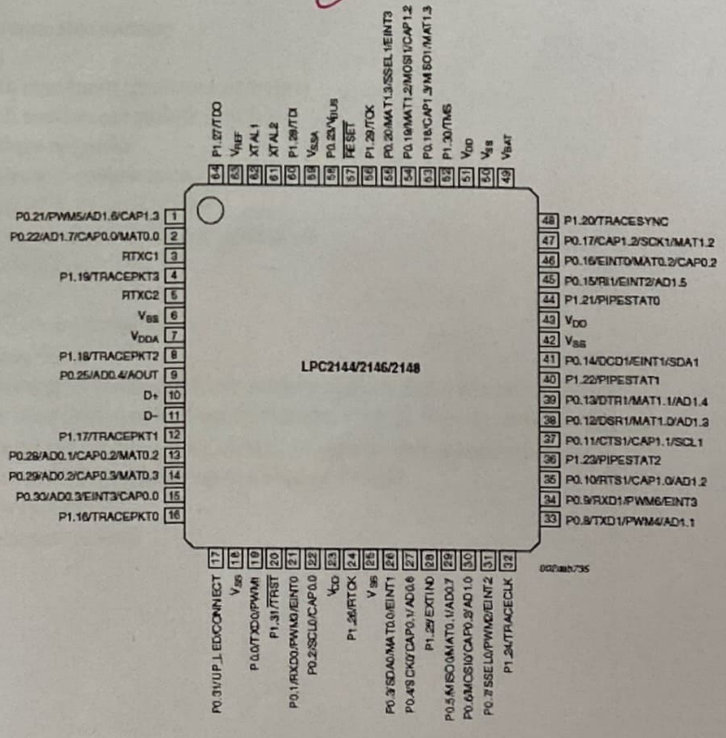
USB - Universal serial bus.

UART - Universal Asynchronous Receiver Transmitter

Architecture of ARM7 (LPC2148)



Pin Diagram



Introduction to PSoC

When developing more complex projects, there is often a need for additional peripheral units, such as operational and instrument amplifiers, filters, timers, digital logic circuits, AD and DA convertors, etc. As a general rule, implementation of the extra peripherals brings in additional difficulties: new components takespace, require additional attention during production of a printed circuit board, increase power consumption...All of these factors can significantly affect the price and development cycle of the project.
The introduction of PSoC microcontrollers has made many engineers' dream come true of having all their project needs covered in one chip.

PSoC: Programmable System on Chip

PSoC (Programmable System on Chip) represents a whole new concept in microcontroller evelopment. In addition to all the standard elements of 8-bit microcontrollers, PSoC chips feature digital and analog Programmable blocks, which themselves allow implementation of large number of peripherals.
Digital blocks consist of smaller programmable blocks that can be configured to allow different developmentoptions. Analog blocks are used for development of analog elements, such as analog filters, comparators, intrumentational (non–) inverting amplifiers, as well as AD and DA convertors.There's a number of different PSoC families you can base your project upon, depending on the projectrequirements. Basic difference between PSoC families is the number of available programmable blocks and the number of input/output pins.Number of components that can be devised is primarily a function of the available programmable blocks.

Depending on the microcontroller family, PSoC chips have 4–16 digital blocks, and 3–12 analog Programmable blocks.

Characteristics of PSoC microcontrollers

Some of the most prominent features of PSoC microcontrollers are:

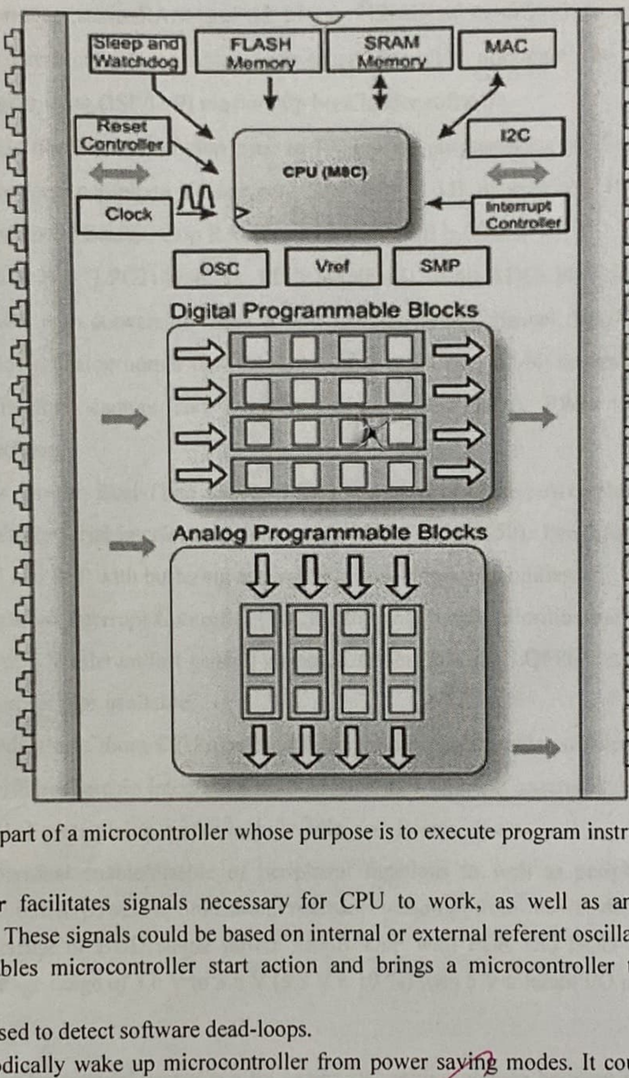
- MAC unit, hardware 8x8 multiplication, with result stored in 32-bit accumulator,
- Changeable working voltage, 3.3V or 5V
- Possibility of small voltage supply, to 1V
- Programmable frequency choice.

Programmable blocks allow you to devise:

- 16K bytes of programmable memory
- 256 bytes of RAM
- AD convertors with maximum resolution af 14 bits
- DA convertors with maximum resolution of 9 bits
- Programmable voltage amplifier
- Programmable filters and comparators
- Timers and counters of 8, 16, and 32 bits
- Pseudorandom sequences and CRC code generators
- Two Full-Duplex UART's
- Multiple SPI devices
- Option for connection on all output pins
- Option for block combining
- Option for programming only the specified memory regions and write protection
- For every pin there is an option of Pull up, Pull down, High Z, Strong, or Open pin state
- Possibility of interrupt generation during change of state on any input/output pin
- I2C Slave or Master and Multi-Master up to speed of 400KH
- Integrated Supervisory Circuit
- Built-in precise voltage reference

System overview

- PSoC microcontrollers are based on 8-bit CISC architecture. Their general structure with basicblocks ispresented in the following image



CPU unit is the main part of a microcontroller whose purpose is to execute program instructions and control workflow of other blocks.
Frequency generator facilitates signals necessary for CPU to work, as well as an array of frequencies thatare used by programmable blocks. These signals could be based on internal or external referent oscillator.
Reset controller enables microcontroller start action and brings a microcontroller to regular state in the caseof irregular events.
Watch Dog timer is used to detect software dead-loops.
Sleep timer can periodically wake up microcontroller from power saving modes. It could be also used as aregular timer.
Input-Output pins enable communication between the CPU unit, digital and analog programmable blocksand outside world.
Digital programmable blocks are used to configure digital programmable components which areselected by user.
Analog programmable blocks are used to configure analog components, like AD and DA converters, filters,and DTMF receivers, programmable, instrumental, inverting, non-inverting and operational amplifiers.
Interrupt controller handles necessary operations in the case of interrupts.
I2C controller Enables hardware realization of an I2C communication.
Voltage reference is vital for the work of analog components that reside inside of analog programmableblocks.
MAC unit is used for operations of hardware signed multiplication of 8-bit numbers.
SMP is a system which can be used as a part of a voltage regulator. For example, it is possible tosupplypower to a PSoC microcontroller from a single 1.5V battery.

Features of ARM DEVELOPMENT KIT Processor:

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- Vectored Interrupt Controller (VIC) with configurable priorities and vector addresses.Up to 45 of 5 V tolerant fast general purpose I/O pins in a tiny LQFP64 package.Up to 21 external interrupt pins available.
- 60MHz maximum CPU clock available from programmable on-chip PLL with settling time of 100μs.On-chip integrated oscillator operates with an external crystal from 1 MHz to 25 MHz.Power saving modes include Idle and Power- down.
- Individual enable/disable of peripheral functions as well as peripheral clock scaling for additional power optimization.Processor wake-up from Power-down mode via external interrupt or BOD.Single power supply chip with POR and BOD circuits:CPU operating voltage range of 3.0 V to 3.6 V (3.3 V ± 10 %) with 5 V tolerant I/O pads.

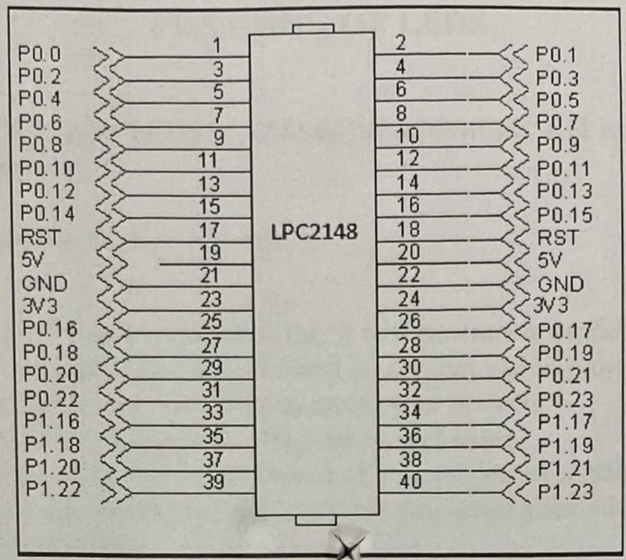
Power Supply:

- The external power can be AC or DC, with a voltage between (9V/12V, 1A output) at 230V AC input.The ARM board produces +5V using an LM7805 voltage regulator, which provides supply to the peripherals.
- LM1117 Fixed +3.3V positive regulator used for processor & processor related peripherals.

Flash Programming Utility

- **NXP (Philips)** - NXP Semiconductors produce a range of Microcontrollers that feature both on-chipFlash memory and the ability to be reprogrammed using In-System Programming technology.

Pin Configuration:



On-board Peripherals:

- 8-Nos. of Point LED's (Digital Outputs)
- 8-Nos. of Digital Inputs (slide switch)
- 2 Lines X 16 Character LCD Display
- I2C Enabled 4 Digit Seven-segment display
- 128x64 Graphical LCD Display
- 4 X 4 Matrix keypad
- Stepper Motor Interface
- 2 Nos. Relay Interface
- Two UART for serial port communication through PC
- Serial EEPROM
- On-chip Real Time Clock with battery backup
- PS/2 Keyboard interface(Optional)
- Temperature Sensor
- Buzzer(Alarm Interface)
- Traffic Light Module(Optional)

RESULT:

Thus the study of ARM processor has been done and ensured its composition with internal features specifically.

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